

Name : UDHAYARAJAN J.
class : MCA - I
Date : 2026-03-22
Roll no : 2202561
Type : CCE - 04
Subject : Research methodology

Research paper:

Title: Impact of Artificial Intelligence on Software Development.

Abstract:

Artificial Intelligence (AI) is a transforming software development by improving productivity, accuracy and automation. This paper examines the impact of AI tools and techniques on modern development practices, highlighting benefits, challenges and future scope.

Introduction:

Software development has evolved rapidly with the integration of AI. Technologies such as machine learning, natural language processing and automation tools are helping developers write, test and deploy code more efficiently. AI is now an essential part of modern software engineering.

Objectives:

- To study the role of AI in software development.
- To analyze the benefits and challenges of AI adoption.
- To understand the future scope of AI in development.

Literature Review:

previous studies show that AI improves code quality, debugging and testing efficiency. AI powered tools assist developers in reducing manual effort and increasing speed. However, concerns about dependency and data privacy are also highlighted.

Research Methodology:

Type of Research: Descriptive research.

Data Source: Secondary data (articles, journals, reports)

Method: Analysis of existing studies and industry practices.

Analysis and Discussion:

Advantages of AI in Software Development:

- Automates code generation and testing.
- Improves accuracy and reduces errors.
- Enhance productivity and efficiency.

Challenges:

- High implementation cost.
- require skilled professionals.
- Risk of over-dependence on AI tools.

Applications:

- Code assistants
- Bug detection and testing automation.
- project management and prediction.

Findings:

- AI significantly reduces development time.
- improves software quality and performance.
- Adoption is increasing in modern IT industries.

Conclusion:

AI is revolutionizing software development by making processes faster, smarter and more efficient. Despite some challenges, it benefits outweigh the limitations, and it will continue to play a major role in future development practices.

References:

- Research journals on AI and software engineering
- Articles from technology websites
- Academic study material

Report type:

Descriptive Research paper